C16 POWER GENERATION
NOT EMISSIONED G-DRIVE
FAILURE & REACTION

First Release, Revision 1.0
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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
AFCRestrLvl1	Air FIlter Clogging first level warning	4B84-1F	Air FIlter	Air FIlter Clogging is detected via a FIlter Clogging Switch connected via Hardware pin. Two warning level are forseen, second level interveens if no action is taken after first level warning, normally reducing engine torque. The measure channel AFCSwt_stDebVal_mp contains the status of the Air filter.	Air filter is clogged	Clean or replace air filter	no lamp (OFF)	19332-31
AFCRestrLvl2	Air FIlter Clogging second level warning	4C84-1F	Air FIlter	Air FIlter Clogging is detected via a FIlter Clogging Switch connected via Hardware pin. Two warning level are forseen, second level interveens if no action is taken after first level warning, normally reducing engine torque. The measure channel AFCSwt_stDebVal_mp contains the status of the Air filter.	Air filter is clogged	Clean or replace air filter	no lamp (OFF)	19588-31
Appdesc_EEP_ERR_CB	ECM problem: EEPROM values for baud rate cannot be read	E16D-1F	ECU EEPROM	Calibrated parameters for baud rate cannot be read from EEPROM	Wrong programming? EEPROM defective?	Reprogram ECM, replace ECM	no lamp (OFF)	57709-31
ARlySCB_0	Short circuit to battery error at high side actuator relay with index x = 0	647D-1F	Actuator Relay	The power stage of relay actuator 0 is monitored by Hardware for electric failures.	Short circuit of wiring to external source or inside connected actuator	Check of wiring or replacement of connected actuator	no lamp (OFF)	25725-31
ARlySCB_1	Short circuit to battery error at high side actuator relay with index x = 1	667D-1F	Actuator Relay	The power stage of relay actuator 1 is monitored by Hardware for electric failures.	Short circuit of wiring to external source or inside connected actuator	Check of wiring or replacement of connected actuator	no lamp (OFF)	26237-31
ARlySCB_2	Short circuit to battery error at high side actuator relay with index x = 2	687D-1F	Actuator Relay	The power stage of relay actuator 2 is monitored by Hardware for electric failures.	Short circuit of wiring to external source or inside connected actuator	Check of wiring or replacement of connected actuator	no lamp (OFF)	26749-31
ARlySCG_0	Short circuit to ground error at high side actuator relay with index x = 0	757D-1F	Actuator Relay	The power stage of Actuator Relay 0 is monitored by Hardware for electric failures.The monitoring for the short circuit to ground is done permanently for CY141 power stages only when the actuator relay is activated.	Short circuit of wiring to ground or inside connected actuator	Check of wiring or replacement of connected actuator	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	30077-31
ARlySCG_1	Short circuit to ground error at high side actuator relay with index x = 1	777D-1F	Actuator Relay	The power stage of Actuator Relay 0 is monitored by Hardware for electric failures.The monitoring for the short circuit to ground is done permanently for CY141 power stages only when the actuator relay is activated.	Short circuit of wiring to ground or inside connected actuator	Check of wiring or replacement of connected actuator	no lamp (OFF)	30589-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
ARlySCG_2	Short circuit to ground error at high side actuator relay with index x = 2	797D-1F	Actuator Relay	The power stage of Actuator Relay 0 is monitored by Hardware for electric failures.The monitoring for the short circuit to ground is done permanently for CY141 power stages only when the actuator relay is activated.	Short circuit of wiring to ground or inside connected actuator	Check of wiring or replacement of connected actuator	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	31101-31
BattUSRCMax	Battery voltage: SRC high for battery voltage sensor	1361-1F	Battery voltage	Static monitoring (signal range check) of the battery voltage.	Alternator voltage governor defect	Replace alternator governor device or alternator	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4961-31
BattUSRCMin	Battery voltage: SRC low for battery voltage sensor	2461-1F	Battery voltage	Static monitoring (signal range check) of the battery voltage.	Battery defect, alternator defect, wiring problems (too high resistance) or ECU defect. Occurrence possible during cold start.	Replace battery or alternator. Check ECU and wiring.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9313-31
BusDiagBusOffErrPasNodeA	CAN Bus: Error passive CAN Bus Node A	911B-1F	CAN Bus	The proper functioning of CAN Bus is monitored. If Bus Off state at CAN controller Node A occurs failure is detected.	Wrong configuration of CAN timing (Prescaler, synchronization jump width, sampling, propagation and phase segments), bad termination resistors, defective network connection and contacts or defective CAN controller with erroneous messages.	Check timing configuration with hardware compatibility, check termination resistors and network connection and contacts. Short circuits to external source, ground or CAN High to Low.Monitor CAN communication for erroneous behaviour of a participant.	no lamp (OFF)	37147-31
BusDiagBusOffNodeA	CAN Bus: Error CAN Bus Node A	951B-1F	CAN Bus	The proper functioning of CAN Bus is monitored. If Bus Off state at CAN controller Node A occurs failure is detected.	Wrong configuration of CAN timing (Prescaler, synchronization jump width, sampling, propagation and phase segments), bad termination resistors, defective network connection and contacts or defective CAN controller with erroneous messages.	Check timing configuration with hardware compatibility, check termination resistors and network connection and contacts. Short circuits to external source, ground or CAN High to Low.Monitor CAN communication for erroneous behaviour of a participant.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	38171-31
CCDiaFOCoHiPres	Fuel in Oil detection: High crankcase pressure detected, first time step	48C3-1F	Crankcase pressure	An increased pressure in the crankcase could damage the engine. Crankcase pressure could be increased because of a clogged blow by filter or if there is a fuel in oil diluition. Different element are examined to discriminate between fuel in oil and other possible problems: - presence of rail errors - crankcase pressure level - oil pressure level	Fuel in oil diluition, or piston ring cracked	Check the oil level; if it is increased there is most probably a leakage of fuel system in the oil swamp. Check rail for leakages, replace oil. Check piston rings (engine power output, abnormal smoke,...)	no lamp (OFF)	18627-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CCDiaFOCoHiPresDelay	Fuel in Oil detection: High crankcase pressure detected , second time step	49C3-1F	Crankcase pressure	An increased pressure in the crankcase could damage the engine. Crankcase pressure could be increased because of a clogged blow by filter or if there is a fuel in oil diluition. Different element are examined to discriminate between fuel in oil and other possible problems: - presence of rail errors - crankcase pressure level - oil pressure level	Fuel in oil diluition, or piston ring cracked	Check the oil level; if it is increased there is most probably a leakage of fuel system in the oil swamp. Check rail for leakages, replace oil. Check piston rings (engine power output, abnormal smoke,...)	no lamp (OFF)	18883-31
CCDiaFOCoOld	Fuel in Oil detection: Fuel in Oil detected since last driving cycle (oil pressure low: low oil viscosity)	4AC3-1F	Crankcase pressure	An increased pressure in the crankcase could damage the engine. Crankcase pressure could be increased because of a clogged blow by filter or if there is a fuel in oil diluition. Different element are examined to discriminate between fuel in oil and other possible problems: - presence of rail errors - crankcase pressure level - oil pressure level	Fuel in oil diluition	Check the oil level; if it is increased there is most probably a leakage of fuel system in the oil swamp. Check rail for leakages, replace oil.	no lamp (OFF)	19139-31
CEngDsTSRCMax	Coolant temp sensor: SRC High for Engine coolant temperature(down stream)	1113-1F	Coolant temp sensor	Electrical Signal range check of Coolant temperature sensor	Sensor defective or short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4371-31
CEngDsTSRCMin	Coolant temp sensor: SRC low for Engine coolant temperature(down stream)	2213-1F	Coolant temp sensor	Electrical Signal range check of Coolant temperature sensor	Sensor defective or short circuit to ground	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	8723-31
CoETSBstPrtTrqLim	Info: Torque Limitation active: Torque limitation caused by turbo charger protection after start	F1EF-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to the VGT protection mainly in high altitude, which was not caused by a failure, but an extreme working condition, which duration is longer than 60.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure concerning the turbocharger is stored in the memory.	Active power reduction > 60.00s due to turbo charger protection	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	61935-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CoETSEngPrtOvhtClnt	Info: Torque Limitation active: Torque limitation caused by high engine coolant temperature	F3A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to high coolant temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if other failures concerning cooling system are present in the failure memory.	Active power reduction > 120.00s due to high engine coolant temperature	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62370-31
CoETSEngPrtOvhtEg	Info: Torque Limitation active: Torque limitation caused by high exhaust gas temperature	F4A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to high Exhaust Gas temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure influencing Exhaust system temperature are present in the failure memory.	Active power reduction > 120.00s due to high Exhaust gas temperature	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62626-31
CoETSEngPrtOvhtFl	Info: Torque Limitation active: Torque limitation caused by high fuel temperature	F5A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to high fuel temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if other failures which could ingluence fuel temperature are present in the failure memory.	Active power reduction > 120.00s due to high fuel temperature	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62882-31
CoETSEngPrtOvhtIntkAir	Info: Torque Limitation active: Torque limitation caused by high air intake temperature	F6A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to high air intake temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if other failures influencing air intake temperature are present in the failure memory.	Active power reduction > 120.00s due to high air intake temperature	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	63138-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CoETSEngPrtOvhtOil	Info: Torque Limitation active: Torque limitation caused by high oil temperature	F7A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to high oil temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if other failures influencing oil temperature are present in the failure memory.	Active power reduction > 120.00s due to high oil temperature	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error. If no other error justifies increased oil temperature and engine is provided with an oil thermostat, check its correctly functioning.	no lamp (OFF)	63394-31
CoETSEngPrtOvhtTrbCh	Info: Torque Limitation active: Torque limitation caused by turbo charger protection	F8A2-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to turbocharger protection after start, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure influencing turbocharger protection are present in the failure memory. Turbocharger protection after start is a functionality which should prevent to stress the turbo until lubrication is optimal after engine start	Active power reduction > 120.00s due to turbocharger protection after start	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	63650-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CoETSEngPrtTrqLim	Info: Torque Limitation active: Torque limitation caused by engine protection	F2EF-1F	Info: Torque Limitation active	The engine is protected against too high air intake gas temperature (VGT protection), to high coolant temperature, too high oil temperature and too high fuel temperature by reducing the torque, if one of the temperatures exceeds the dedicated limit: The maximum engine speed is limited to protect the VGT to run without lube oil after cold start. The Torque is limited depending on the difference between actual engine speed and the allowed speed which depends on engine temperature and time since start (from 655.00mg/hub at 50.00rpm over 655.00mg/hub at 2000.00rpm to 655.00mg/hub at 6000.00rpm). Torque is reduced depending on oil temperature value according to a calibration curve. Torque is reduced depending on coolant temperature (by factor 1.50- at -40.14°C over 1.50- at -20.14°C and 1.50- at -10.14°C and 1.50- at -0.14°C and 1.50- at 9.86°C Torque is reduced depending on fuel temperature (by factor 1.50- at -40.14°C over 1.50- at -20.14°C and 1.50- at -10.14°C and 1.50- at -0.14°C and 1.50- at 9.86°C If calculated exhaust temperature exceeds 739.96°C the torque is reduced dependant on engine speed and actual air mass flow. If the reduction is > 15.00% for more than 120.00s, a failure is stored as confirmation that there was an active performance limitation, which was not caused by a failure, but an extreme working condition . This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure concerning the engine protection/engine speed is stored in the memory.	Active power reduction > 120.00s due to engine mechanics protection	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62191-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CoETSInjSysTrqLim	Info: Torque Limitation active: Torque limitation caused by injection system	F3EF-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to the high pressure fuel pump protection against too high fuel temperature, which was not caused by a failure, but an extreme working condition, which duration is longer than 120.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure concerning the injection system/fuel temperature is stored in the memory.	Active power reduction > 120.00s due to high pressure fuel pump protection	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62447-31
CoETSLimInfo	Info: One or more torque limitations active	FAEF-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation	One or more torque reductions are present	No actions necessary due to this failure alone. If powerreduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	64239-31
CoETSNTCTrqLim	Info: Torque Limitation active: Torque limitation caused by engine brake	F4EF-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation after engine brake release, which was not caused by a failure, but an extreme working condition, which duration is longer than 60.00s. This failure is thought to have a confirmation in case the driver complained a temporary "power lack". This failure does not ask for any action except, to check if another failure concerning the engine break is stored in the memory.	Active power reduction > 60.00s due to torque rise limitation after engine brake release.	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	62703-31
CoETSPrfImTrqLim	Info: Torque Limitation active: Torque limitation caused by OBD performance limiter	F6EF-1F	Info: Torque Limitation active	This failure path is only for information, that there was an active performance limitation due to OBD legislation, which was caused by too high NOx emissions, which duration is longer than 60.00s. This failure does not ask for any action except, to check if another failure concerning the NOx emissions is stored in the memory (with FID_CoVehPrfmLim%%OBD with %%=11,12,21 or 22)	Active power reduction due to the OBD performance limiter because of too high NOx emissions	No actions necessary due to this failure alone. If powerreduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	63215-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CoETSSCRPrtTrqLim	Info: Torque Limitation active: Torque limitation caused by SCRCAT protection	F7EF-1F	Info: Torque Limitation active	If the SCR temperature exceeds 999.96°C > 0.30 sec the max torque is limited by a factor, which depends by how much the 999.96°C are exceeded to protect the SCR for thermal ageing. If the power reduction exceeds 15.00% for more than 60.00s , a failure path is activated for information, that there was really an active performance limitation due to the SCR protection, which was not caused by a failure, but an extreme working condition. This failure is thought to have a confirmation in case the driver complained a temporary "power lack" without any information. This failure does not ask for any action except, to check if another failure is stored in the memory which could lead to too high exhaust temperatures.	Active power reduction due to excessive temperatures in the SCR catalyst	No actions necessary due to this failure alone. If power reduction occurred due to actual defect, the failure triggering the torque limitation should also be in the failure memory. Follow troubleshooting of this root error.	no lamp (OFF)	63471-31
ComTOTSC1PEAct	CAN Bus: Active DFC TimeOut of Torque Speed Control TSC1PE Message	98BB-1F	CAN Bus	The message TSC1PE contains the information for torque/speed demand from PTO Controller to EDC. TSC1 CAN messages (Torque/Speed Control) are sent at fast rate if Torque/speed request is active and at slower rate if no request is active, to reduce CAN traffic, so different timeout monitor are provided: a fast one if a request is active (active timeout), and a slow one if no request is active (passive timeout).	Defective CAN controller of PTO Controller, CAN cable disconnected or broken. Short circuit in wiring.	Check presence and correct connection of the PTO Controller to the network, Check correct functioning of the CAN controller. Check wiring.	no lamp (OFF)	39099-31
ComTOTSC1TEAct	CAN Bus: Active Time out for Torque Speed Control TSC1TE message	9CBB-1F	CAN Bus	The message TSC1TE contains the information for torque/speed demand from Transmission Controller (AMT) to EDC. TSC1 CAN messages (Torque/Speed Control) are sent at fast rate if Torque/speed request is active and at slower rate if no request is active, to reduce CAN traffic, so different timeout monitor are provided: a fast one if a request is active (active timeout), and a slow one if no request is active (passive timeout).	Defective CAN controller of Transmission Controller, CAN cable disconnected or broken. Short circuit in wiring.	Check presence and correct connection of the Transmission Controller to the network, Check correct functioning of the CAN controller. Check wiring.	no lamp (OFF)	40123-31
CooltP1Lo	Primary Coolant pressure: value too low	2D73-1F	Coolant pressure	Primary coolant pressure in the cooling circuit is monitored Primary coolant pressure sensor is connected to ECU via hardware pin. Sea water pressure raw voltage value measured by sensor read from the ADC is converted to physical value. The physical range check is performed to assure that the sensed physical value is within a "sensible" range, allowed for that sensor.	Primary coolant pressure too low	Sensor drifted? FIlter clogged? Pump defective Restriction pump up stream	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	11635-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
CrCsPPhysRngLo	Crankcase differential pressure : Physical signal below lower limit (physical SRC)	22A3-1F	Crankcase pressure	The raw signal measured by crankcase differential pressure sensor is converted into a physical value. Measured physical value should lie in a defined range.	Excessively low crankcase differential pressure is measured	Check cause for value to fall below minimum allowed threshold (like environmental factors, system problem or engine misuse). If threshold exceeding is implausible check wiring and the accuracy of the sensor and replace it if necessary.	no lamp (OFF)	8867-31
CrCsPSRCMax	Crankcase differential pressure : Signal Range Check high	13A3-1F	Crankcase pressure	The electrical Signal of the Crankcase differential pressure sensor is monitored. In case of defect recognition a substitute value is taken.	Short circuit of sensor to external source or defective sensor	Check wiring and sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5027-31
CrCsPSRCMin	Crankcase differential pressure : Signal Range Check low	24A3-1F	Crankcase pressure	The electrical Signal of the Crankcase differential pressure sensor is monitored. In case of defect recognition a substitute value is taken.	Short circuit of sensor to ground or defective sensor	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9379-31
EEPEraseErr	ECU EEPROM: EEP sector erase failed	E26D-1F	ECU EEPROM	A functionality for reading writing and erasing values to a memory media (Flash or EEPROM) is available. These operations are monitored and if problems arise a failure is reported.	Wrong programming/flashing of the ECM, internal defect of the ECM.	Try to flash the ECM correctly with a proper dataset, if defect persists: replace ECM.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57965-31
EEPRdErr	ECU EEPROM: EEP Read Error based on the error for more blocks	E36D-1F	ECU EEPROM	A functionality for reading writing and erasing values to a memory media (Flash or EEPROM) is available. These operations are monitored and if problems arise a failure is reported.	Wrong programming/flashing of the ECM, internal defect of the ECM.	Try to flash the ECM correctly with a proper dataset, if defect persists: replace ECM.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58221-31
EEPWrErr	ECU EEPROM: EEP Write Error based on the error for one block	E46D-1F	ECU EEPROM	A functionality for reading writing and erasing values to a memory media (Flash or EEPROM) is available. These operations are monitored and if problems arise a failure is reported.	Wrong programming/flashing of the ECM, internal defect of the ECM.	Try to flash the ECM correctly with a proper dataset, if defect persists: replace ECM.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58477-31

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EGTCondNOxEst	SCR NOx calculation: multi signal defects in NOx estimation	3BDA-1F	SCR NOx calculation	The NOx Estimation (= calculated engine raw emission based on several input data's) is essential for the calculation of the needed AdBlue dosing quantity on the SCR system and for controlling the NOx emissions. If the calculation is not reliable due to a defective sensor, the NOx control is no more correct. Therefore this failure path is set by the FId_EGTCondGsCnstn, which will be activated by a failure due to which the NOx estimation is no more reliable.	see leading failure (failure also stored in memory, which activates FId_EGTCondGsCnstn)	see failure stored in memory, which activated FId_EGTCondGsCnstn	no lamp (OFF)	15322-31
EngDaTPrevWarn	Info: Engine temperature exceeded pre-warn level	F523-1F	Info: Torque Limitation active	Two engine temperature thresholds are available to signal high engine temperature. Pre-warning is set if engine temperature > 101.96°C Warning is set if engine temperature > 105.96°C	Too high engine temperature	Cooling system problem or simply overheated engine due to heavy use or high environmental temperature	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62755-31
EngDaTWarn	Info: Engine temperature exceeded warn level	F623-1F	Info: Torque Limitation active	Two engine temperature thresholds are available to signal high engine temperature. Pre-warning is set if engine temperature > 101.96°C Warning is set if engine temperature > 105.96°C	Too high engine temperature	Cooling system problem or simply overheated engine due to heavy use or high environmental temperature	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	63011-31
EngICO	ECU internal Check: Injection cut off demand (ICO) for shut off coordinator	E536-1F	ECU internal Check	Injection cut off is the request of ECM to inhibit fuel injection to limit the engine speed to a controllable range. This procedure is triggered when some ECM internal failures which concern reliability of accelerator pedal signal , fuel injection parameters or engine speed are detected.	Another failure requested engine speed limitation. This failure path should always appear in failure memory with another lead failure from ECM monitoring level (level 2)	No action, this is only an information path	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58678-31
EngPrtOvrSpd	Engine Overspeed: Overspeed detection in component engine protection	F1A2-1F	Engine Overspeed	Engine speed is monitored and in case of overspeed detection a failure is reported. Unacceptably high engine speed may arise in the event of an error in the ECM or under certain operating states like downhill travel, in this case the failure should does not indicate an error but simply informs about a "misuse" of the engine.	Engine overspeed has occurred	No reactions necessary only if this fault/information status reoccurs frequently. In this case check driving conditions of vehicle, engine speed acquisition and injection system for quantity set point and actual value during fault recognition, check also for ECM errors.	no lamp (OFF)	61858-31
EnvPSRCMax	Ambient pressure sensor: SRC High for Environment Pressure	13C4-1F	Ambient pressure sensor	Environmental pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through Analogic to Digital Converter and transformed to a physical value in EnvP_pSens. A signal range check is performed on the signal. Sensor is inside ECU.	Ambient pressure sensor inside ECU is defect	Replace ECU	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5060-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
EnvPSRCMin	Ambient pressure sensor: SRC low for Environment Pressure	24C4-1F	Ambient pressure sensor	Environmental pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through Analogic to Digital Converter and transformed to a physical value in EnvP_pSens. A signal range check is performed on the signal. Sensor is inside ECU.	Ambient pressure sensor inside ECU is defect	Replace ECU	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9412-31
EpmCaSI1ErrSig	Engine speed sensing: DFC for camshaft signal diagnosis - disturbed signal	4112-1F	Engine speed sensing	The Engine Position Management module is responsible for signals from camshaft and crankshaft sensor evaluation: angle and speed determination, signal plausibility and quality are monitored by this module. Plausibility of number and position of the camshaft edges detected by camshaft speed sensor is monitored here.	Cam Shaft speed signal not stable: - electrical intermittent open circuit - sensor not fixed correctly - electrical disturbances due to damaged isolation of wiring harness - electrical disturbances due to special electrical features	Check wiring, sensor installation and proper functioning of camshaft phase sensor (evaluate raw signals) Check if sensor fixed correctly (should not be moveable),check wiring harness and contacts, check if electrical lines with high electric performance are additionally installed (components not validated by the vehicle manufacturer).	no lamp (OFF)	16658-31
EpmCaSI1NoSig	Engine speed sensing: DFC for camshaft signal diagnosis - no signal	4212-1F	Engine speed sensing	The Engine Position Management module is responsible for signals from camshaft and crankshaft sensor evaluation: angle and speed determination, signal plausibility and quality are monitored by this module. Loss of camshaft speed sensor signal is monitored here.	Camshaft speed sensor defective Wiring harness "open circuit" Camshaft sensor not mounted	Check if sensor mounted correctly Check wiring harness Check sensor functionality and replace it if necessary	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	16914-31
EpmCaSI1OfsErr	Engine speed sensing: DFC for camshaft offset angle exceeded	3312-1F	Engine speed sensing	The Engine Position Management module is responsible for signals from camshaft and crankshaft sensor evaluation: angle and speed determination, signal plausibility and quality are monitored by this module. Plausibility between camshaft and crankshaft speed sensor signal is monitored here.	Camshaft sensor position deviates form crankshaft position: - camshaft mal positioned (e.g. tooth belt: 1 tooth skipped) - camshaft position wheel mal positioned (if adjustable) - camshaft sensor position mal positioned (if adjustable) - crankshaft signal mal positioned (same as camshaft)	Check correct positions of camshaft signal wheel and crankshaft signal wheel to piston position	no lamp (OFF)	13074-31
EpmCrSErrSig	Engine speed sensing: Crankshaft signal disturbance	4412-1F	Engine speed sensing	The Engine Position Management module is responsible for signals from camshaft and crankshaft sensor evaluation: angle and speed determination, signal plausibility and quality are monitored by this module. Plausibility of crankshaft speed signal is monitored here.	- High vibration of crank shaft signal sensor - Intermittent open circuit of sensor - Electrical disturbances due to damaged isolation of wiring harness - Electrical disturbances due external electromagnetic disturbances	Check if sensor fixed correctly (should not be moveable) Check wiring harness and contacts Check if electrical lines with high electric performance are additionally installed (components not validated by the vehicle manufacturer)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	17426-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
EpmCrSNoSig	Engine speed sensing: Crankshaft signal loss	4512-1F	Engine speed sensing	The Engine Position Management module is responsible for signals from camshaft and crankshaft sensor evaluation: angle and speed determination, signal plausibility and quality are monitored by this module. Presence of crankshaft speed signal is monitored here.	Crank shaft speed sensor defect Crank shaft speed sensor not mounted correctly Crank shaft speed sensor not connected electrically, open circuit or short circuit.	Check if crank shaft speed sensor mounted correctly Check wiring harness to open/short circuit Check crank shaft speed sensor for correct function and replace it if necessary	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	17682-31
ExhMgTHcMonWarn	HC accumulated in the exhaust reached second warning level	31E9-1F	None	This function realizes a component protection for the exhaust system in case of using the vehicle for a long time in idle speed and with cold environmental temperatures. This causes the system to accumulate a lot of liquid HC mass in the exhaust system which could burn up when engine is driven under load again and cause exhaust system to catch fire or even to destroy the vehicle. Threshold for fault detection should be applied so that a warning or an engine shut down is triggered before HC accumulation becomes dangerous.	Engine driven a long time at idle condition (low temperatures)	Warm up engine and exhaust by driving under load	no lamp (OFF)	12777-31
FlFilClogPreFil	Fuel Filter: Pre filter is clogged	31E5-1F	Fuel Filter	The functionality detects fuel prefilter clogging evaluating fuel pressure after pre filter	Fuel pre filter is clogged	Clean or replace fuel pre filter	no lamp (OFF)	12773-31
FlFltCtlClogDet	Fuel Filter is clogged (sensor switch active)	4BB5-1F	Fuel Filter	This function evaluates the fuel filter clog detection switch. The switch evaluates the differential pressure over the fuel filter. If that exceeds a certain value, the switch is operated and indicates a clogged fuel filter. Digital signal evaluation is processed by this function. Additionally an event counter is implemented which counts how many times the fuel filter has suffered clogging.	Fuel filter clogged	CLean or replace fuel filter	no lamp (OFF)	19381-31
FlFltCtlClogDetPlaus	Fuel FIlter Clog Switch blocked in clogged status	4CB5-1F	Fuel Filter	This function evaluates the fuel filter clog detection switch. The switch evaluates the differential pressure over the fuel filter. If that exceeds a certain value, the switch is operated and indicates a clogged fuel filter. Digital signal evaluation is processed by this function. Additionally an event counter is implemented which counts how many times the fuel filter has suffered clogging.	Defective filter clogging switch, blocked in active position	Check and replace filter clog switch	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	19637-31
FlFltHtOL	Fuel filter heater: No load error in power stage of fuel filter heating	51B5-1F	Fuel filter heater	The Power stage of Fuel Filter heater is monitored by Hardware for electric failures.	Broken or disconnected wiring or defective fuel filter heater.	Check of wiring or replace fuel filter heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	20917-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
FlFltHtOvrTemp	Fuel filter heater: Over Temperature error at ECM power stage	82B5-1F	Fuel filter heater	The Power stage of Fuel Filter heater is monitored by Hardware for electric failures.	High battery voltage, high temperature inside ECU, high load or wiring problem	Check load and output, check wiring, replace ECU	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	33461-31
FlFltHtSCB	Fuel filter heater: Short circuit to battery error in power stage of fuel filter heating	63B5-1F	Fuel filter heater	The Power stage of Fuel Filter heater is monitored by Hardware for electric failures.	Short circuit of wiring to external source or inside fuel filter heater	Check of wiring, replace fuel filter heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25525-31
FlFltHtSCG	Fuel filter heater: Short circuit to ground error in power stage of fuel filter heating	74B5-1F	Fuel filter heater	The Power stage of Fuel Filter heater is monitored by Hardware for electric failures.	Short circuit of wiring to ground or inside fuel filter heater	Check of wiring, replace fuel filter heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	29877-31
FlSys_WtDet	Water in Fuel Determination: Water in Fuel	4415-1F	Water in Fuel determination	The Water in fuel detection is carried out with a digital sensor placed in fuel filter	Water in fuel filter	Clean fuel filter Check Sensor measured value	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	17429-31
FlSys_WtDet_SensChk	Water in fuel Sensor defective	4F15-1F	Water in Fuel determination	At key-on, the fuel filter electronics performs a self-test, setting the WaterInFuel signal to the alarm condition for a certain time Ttest (Ttest = 3 sec. ± 20 %) to perform a connection check.	Water in Fuel Sensor defective	Check and replace Water in Fuel Sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	20245-31
FOSensNotPlaus	Crankcase pressure: value not plausible at engine start	3DA3-1F	Crankcase pressure	When oil level is increased too much because of fuel in oil diluition, pressure in the crankcase is increased and eventually overcomes the tightness of the blow-by filter. Plausibility of crankcase pressure at start is checked here.	Crankcase pressure sensor value not plausible	Check crankcase pressure sensor	no lamp (OFF)	15779-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
FuelTPhysRngHi	Fuel Temperature Sensor: Physical Signal above maximum limit	1A15-1F	Fuel Temp Sensor	Measured Fuel temperature value should lie in a defined range.	Excessive Fuel temperature measured	Check cause for exceeding maximum allowed threshold (like environmental factors, system problem or engine misuse). If threshold exceeding is implausible check wiring and the accuracy of the sensor and replace it if necessary.	no lamp (OFF)	6677-31
FuelTSRCMax	Fuel Temp Sensor: SRC high for fuel temperature sensor	1C15-1F	Fuel Temp Sensor	The electrical Signal of the Fuel temperature sensor is monitored. In case of defect recognition a default value for the fuel temperature is taken. In case of defect a fix value or the actual engine temperature can be taken as substitution, depending on configuration (by data set) and reliability of the sensors.	Sensor defective or short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	7189-31
FuelTSRCMin	Fuel Temp Sensor: SRC low for fuel temperature sensor	2D15-1F	Fuel Temp Sensor	The electrical Signal of the Fuel temperature sensor is monitored. In case of defect recognition a default value for the fuel temperature is taken. In case of defect a fix value or the actual engine temperature can be taken as substitution, depending on configuration (by data set) and reliability of the sensors.	Sensor defective or short circuit to ground	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	11541-31
GptaEEPData1BlkErr	ECU EEPROM: Error in block EEPData1	E56D-1F	ECU EEPROM	Error in EEPROM block EEPData1. Correction values not be calculated. Error normally disabled on serial application, only relevant for Bosch during calibration activity.	Wrong programming/flashing of the ECM, internal defect of the ECM.	Reprogram ECM, replace ECM	no lamp (OFF)	58733-31
IAirHtOL_0	Air heater: Open load on Power stage 0	5614-1F	Air heater	The Power stage of Intake Air Heater actuator 0 (Digital Output) is monitored by Hardware for electric failures.	Broken or disconnected wiring or defective air heater	Check of wiring or replace air heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	22036-31
IAirHtOvrTemp_0	Air heater: Over temperature on Power stage 0	8814-1F	Air heater	The Power stage of Intake Air Heater actuator 0 (Digital Output) is monitored by Hardware for electric failures.	High battery voltage, high temperature inside ECU, high load or wiring problem	Check load and output, check wiring, replace ECU	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	34836-31
IAirHtSCB_0	Air heater: Short circuit to battery on power stage 0	6A14-1F	Air heater	The Power stage of Intake Air Heater actuator 0 (Digital Output) is monitored by Hardware for electric failures.	Short circuit of wiring to external source or inside air heater	Check of wiring, replace air heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	27156-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IAirHtSCG_0	Air heater: Short circuit to ground error on power stage 0	7C14-1F	Air heater	The Power stage of Intake Air Heater actuator 0 (Digital Output) is monitored by Hardware for electric failures.	Short circuit of wiring to ground or inside air heater	Check of wiring, replace air heater	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	31764-31
InjCrvInjLimChrgBal	Injector: Number of injections is limited by charge balance of booster capacity	1636-1F	Injector	The compatibility between requested number of injections and current capability of the injection system of fulfilling that request is checked. Number of requested number of injections is beyond current booster capacity.	Application of number of requested injections not compatible with system capability or charging of injector booster insufficient	Check battery voltage, check power supply of ECU, if both OK replace ECU	no lamp (OFF)	5686-31
InjCrvInjLimSys	Injector: Number of injections is limited by system	F836-1F	Injector	The compatibility between requested number of injections and current capability of the injection system of fulfilling that request is checked. Number of requested number of injections is beyond maximal number of injections released by application for the system.	Dataset application wrong, too high number of injections allowed.	Dataset should be replaced, contact Help Desk	no lamp (OFF)	63542-31
InjCrvNumInjRtmLim	Injector: Number of injections is limited by runtime	F936-1F	Injector	The compatibility between requested number of injections and current capability of the injection system of fulfilling that request is checked. Number of requested number of injections is beyond maximal number of injections released for the current engine speed. At high speed the number of possible injections is limited by computing time needed for calculations.	Dataset application wrong, too high number of injections allowed.	Dataset should be replaced, contact Help Desk	no lamp (OFF)	63798-31
InjVlvPresMin	Injector: check of minimum rail pressure	2B36-1F	Injector	The minimum rail pressure value necessary to allow fuel injection is monitored.	Rail pressure controller problem (metering unit sticking), High Pressure pump efficiency too low, no fuel delivery from gear pump to high pressure pum or system leakages	Check presence of other more specific failures related to rail pressure controller. Check high pressure pump delivery, metering unit, check for rail or injector leakages. Check for clogged main fuel filter	no lamp (OFF)	11062-31
IVAdjDiaIVAdj_0	Injector Adjustment: IMA programming not correct for injector cyl 1	F176-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	61814-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IVAdjDiaIVAdj_1	Injector Adjustment: IMA programming not correct for injector cyl 4	F276-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62070-31
IVAdjDiaIVAdj_2	Injector Adjustment: IMA programming not correct for injector cyl 2	F376-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62326-31
IVAdjDiaIVAdj_3	Injector Adjustment: IMA programming not correct for injector cyl 6	F476-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62582-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IVAdjDiaIVAdj_4	Injector Adjustment: IMA programming not correct for injector cyl 3	F576-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62838-31
IVAdjDiaIVAdj_5	Injector Adjustment: IMA programming not correct for injector cyl 5	F676-1F	Injector Adjustment	The injectors are corrected individually (IMA Code) to guarantee exact fuel injection quantity of each injector from lowest to highest fuel quantity. The correction is done at the end of production line (EOL) and after substitution of the injectors with the tester. The data's are stored in the EEPROM. The stored values in the EEPROM are protected with checksum parameters. At each Key 15 "on" the correction data's are read from the EEPROM and checked, if the values are plausible and the checksums are correct. If not, the injector will not be corrected and a failure will be stored.	Invalid IMA code has been programmed wrong programming/flashing of the EDC (initialization of EEPROM) internal defect of the ECU	Reprogram IMA code Reflash the ECU (including EEPROM initialisation) if defect persists: replace EDC	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	63094-31
IVDiaBnkShCir_0	Injector: bank0, short circuit	7116-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (over current, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected. Two banks are present.	Short circuit in wiring or injector connected to bank 0	Check the wiring and injectors connected to bank 0.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	28950-31
IVDiaBnkShCir_1	Injector: bank1, short circuit	7216-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (over current, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected. Two banks are present.	Short circuit in wiring or injector connected to bank 1.	Check the wiring and injectors.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	29206-31

Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IVDiaChp	Injector: ECM internal failure of Injector actuation (chip error)	E126-1F	ECU Internal	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Defective ECM	Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57638-31
IVDiaCylNoLd_0	Injector: open load, injector cylinder 1	5316-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 1 is detected.	Check wiring, replace fuel injector of cylinder 1.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	21270-31
IVDiaCylNoLd_1	Injector: open load, injector cylinder 4	5416-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 4 is detected.	Check wiring, replace fuel injector of cylinder 4.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	21526-31
IVDiaCylNoLd_2	Injector: open load, injector cylinder 2	5516-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 2 is detected.	Check wiring, replace fuel injector of cylinder 2.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	21782-31
IVDiaCylNoLd_3	Injector: open load, injector cylinder 6	5616-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 6 is detected.	Check wiring, replace fuel injector of cylinder 6.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	22038-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IVDiaCylNoLd_4	Injector: open load, injector cylinder 3	5716-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 3 is detected.	Check wiring, replace fuel injector of cylinder 3.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	22294-31
IVDiaCylNoLd_5	Injector: open load, injector cylinder 5	5816-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Open load for injector of cylinder 5 is detected.	Check wiring, replace fuel injector of cylinder 5.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	22550-31
IVDiaCylShCir_0	Injector: short circuit, injector cylinder 1	6226-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 1.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25126-31
IVDiaCylShCir_1	Injector: short circuit, injector cylinder 4	6326-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 4.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25382-31
IVDiaCylShCir_2	Injector: short circuit, injector cylinder 2	6426-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 2.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25638-31

Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
IVDiaCylShCir_3	Injector: short circuit, injector cylinder 6	6526-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 6.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25894-31
IVDiaCylShCir_4	Injector: short circuit, injector cylinder 3	6626-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 3.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	26150-31
IVDiaCylShCir_5	Injector: short circuit, injector cylinder 5	6726-1F	Injector	Hardware errors in the injectors and respective ECM power stages are investigated. Diagnostic procedure uses pattern detection to identify specific errors and trigger appropriate substitute reaction. The behaviour pattern describes in a bit mask different detected problem (overcurrent, command collision, RAM error,short circuit,....) when the expected combination of errors is detected (measured and reference pattern are matching) a failure is detected.	Short circuit in wiring or injector	Check wiring, replace fuel injector of cylinder 5.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	26406-31
IVDiaSPITout	ECU Internal: SPI communication timeout	E336-1F	ECU Internal	SPI bus communication is monitored for time out	Internal ECU error	Replace ECU	no lamp (OFF)	58166-31
MaxPTOSwt	PTO actuation: Defect Fault Check for signal value above maximum limit	11A1-1F	PTO actuation	A range and plausibility check for the value of PTO switch are performed. The source for the multi value PTO switch is an analogic input.	PTO switch defective (possible short circuit)	Check PTO switch and wiring	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4513-31
MeUnOL	Fuel metering unit: open load of metering unit output	5255-1F	Fuel metering unit	The power stage of the fuel metering unit to common rail (MeUn) is monitored electrically for voltage and current by Hardware.	Broken or disconnected wiring, defective metering unit	Check of wiring or metering unit	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	21077-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MeUnOT	Fuel metering unit: over temperature of device driver of metering unit	8355-1F	Fuel metering unit	The power stage of the fuel metering unit to common rail (MeUn) is monitored electrically for voltage and current by Hardware.	High battery voltage, high temperature inside ECU, high load or wiring problem	Check of wiring or metering unit. Check resistance of wiring or metering unit. Check for ECU cooling	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	33621-31
MeUnShCirHSBatt	Fuel Metering Unit: Short Circuit to Battery at High Side	6455-1F	Fuel metering unit	The Power stage of Metering Unit at High Side is monitored by Hardware for electric failures.	Short circuit of wiring to battery or inside Metering Unit at High Side	Check of wiring, replace Metering Unit	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	25685-31
MeUnShCirHSGnd	Fuel Metering Unit: Short Circuit to Ground at High Side	7555-1F	Fuel metering unit	The Power stage of Metering Unit at High Side is monitored by Hardware for electric failures.	Short circuit of wiring to ground or inside Metering Unit at High Side	Check of wiring, replace Metering Unit	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	30037-31
MeUnShCirLSBatt	Fuel Metering Unit: Short Circuit to Battery at Low Side	6655-1F	Fuel metering unit	The Power stage of Metering Unit at Low Side is monitored by Hardware for electric failures.	Short circuit of wiring to battery or inside Metering Unit at Low Side	Check of wiring, replace Metering Unit	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	26197-31
MeUnShCirLSGnd	Fuel Metering Unit: Short Circuit to Ground at Low Side	7755-1F	Fuel metering unit	The Power stage of Metering Unit at Low Side is monitored by Hardware for electric failures.	Short circuit of wiring to ground or inside Metering Unit at Low Side	Check of wiring, replace Metering Unit	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	30549-31
MoCADCNTP	ECU Internal: Diagnostic fault check to report the ADC no-load test pulse error	E21D-1F	ECU Internal	The Analogic Digital Converter (ADC) of the ECM is monitored for plausibility using redundantly read input (accelerator pedal) and test voltage inputs.	ECM internal failure	Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57885-31
MoCADCTst	ECU Internal: Diagnostic fault check to report the ADC test error	E31D-1F	ECU Internal	The Analogic Digital Converter (ADC) of the ECM is monitored for plausibility using redundantly read input (accelerator pedal) and test voltage inputs.	ECM internal failure	Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58141-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoCADCVltgRatio	ECU Internal: Diagnostic fault check to report the error in Voltage ratio in ADC monitoring	E41D-1F	ECU Internal	The analogic Digital Converter (ADC) of the ECM is monitored for plausibility using redundantly read input (accelerator pedal) and test voltage inputs.	ECM internal failure	Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58397-31
MoCComErrCnt	ECU Internal: Diagnostic fault check to report errors in query-/response-communication	E51D-1F	ECU Internal	Communication between ECM processor and power stage controller over SPI bus is monitored. Error is detected if no response or an erroneous response is detected by the processor.	Defective monitoring module or CPU (e.g. impaired functioning of the CPU clock) of the EDC. Disturbed SPI-Bus.	If error exists only temporary (i.e. injection reoccurs) error can be ignored and error deleted in the fault memory. In case of a permanent error the injection remains blocked and the ECU has to be replaced	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58653-31
MoCComSPI	ECU Internal: Diagnostic fault check to report errors in SPI-communication	E61D-1F	ECU Internal	Communication between ECM processor and power stage controller over SPI bus is monitored. Error is detected if no response or an erroneous response is detected by the processor.	Disturbed SPI-Bus	If ECM functions correctly after recovery only temporary SPI error occurred and fault memory can be deleted and error ignored. If permanent SPI error is present and the ECM does not leave boot block --> Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58909-31
MoCROMErrXPg	ECU Internal: Diagnostic fault check to report multiple error while checking the complete ROM-memory	E71D-1F	ECU Internal	The complete ROM memory is checked once per driving cycle, at each ECM shut off phase (after-run).	Corrupted datas in the ECM,ECM internal failure	Reprogram ECM, replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59165-31
MoCSOPErrMMRespByte	ECU Internal Check: Communication timeout from CPU to Monitoring Module	E13D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57661-31
MoCSOPErrNoChk	ECU Internal Check: Error during Shut Off Path test, uncertain cause (defective injector or shut-off path)	E23D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure or injector problem	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57917-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoCSOPErrRespTime	ECU Internal Check: Wrong set response time	E33D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58173-31
MoCSOPErrSPI	ECU Internal Check: Too many SPI errors during Shut Off Path test execution.	E43D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58429-31
MoCSOPLoLi	ECU Internal: Diagnostic fault check to report the error in undervoltage monitoring	E53D-1F	ECU Internal	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58685-31
MoCSOPMM	ECU Internal: Diagnostic fault check to report that line to power stage supply is not working correct	E63D-1F	ECU Internal	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58941-31
MoCSOPOSTimeOut	ECU Internal Check: OS timeout in the shut off path test. Failure setting the alarm task period.	E73D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM need to be replaced, contact nearest head office for authorisation.	Contact Help Desk	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59197-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoCSOPPsvTstErr	ECU Internal Check: Error during positive test	E83D-1F	ECU internal Check	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59453-31
MoCSOPTimeOut	ECU Internal: Diagnostic fault check to report the timeout in the shut off path test	E69D-1F	ECU Internal	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59037-31
MoCSOPUpLi	ECU Internal: Diagnostic fault check to report the error in overvoltage monitoring	E79D-1F	ECU Internal	The ECM comprises so-called redundant shut-off paths which deactivate all power stages relevant to injection if certain internal ECU errors are detected. In order to test the correct functioning of the shut off procedure every path is shortly activated during ECM initialization and following this the fuel injection power stage is activated. Since, however, the Shut off is active no injection should occur if procedure functions correctly.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59293-31
MoFAPP	ECU Internal: Diagnostic fault check to report the accelerator pedal position error	E5B1-1F	ECU Internal	A redundant acquisition and plausibility check of accelerator pedal signal for function monitoring is provided in the ECM at software level 2 (monitoring level). The ECM is configured for a potentiometer/switch accelerator pedal.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	58801-31
MoFESpd	ECU Internal: Diagnostic fault check to report the engine speed error	E812-1F	ECU Internal	A redundant check for engine speed is performed for function monitoring at software level 2. Monitoring in the ECM takes place in 3 levels: Level 1: Functions level (engine control, component monitoring) Level 2: Function monitoring level (detects defective processes of functions in level 1) Level 3: Controller monitoring level (check corrects processing of ECM program commands)	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	59410-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoFInjDatET	ECU Internal Check: Error in the plausibility of the injection energizing time	EC36-1F	ECU internal Check	Energising time request plausibility for the injectors is monitored by ECM monitoring software (level 2). ECM level 2 monitors ECM level 1 software (user software).	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	60470-31
MoFInjDatPhi	ECU Internal: Error in the plausibility of the start of energising angles	ED36-1F	ECU Internal	Start of energising time for the injectors, is monitored for plausibility by ECM monitoring software (level 2). ECM level 2 monitors ECM level 1 software (user software).	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	60726-31
MoFInjQnt	ECU Internal: Diagnostic fault check to report the error due to non plausibility in ZFC	EE36-1F	ECU Internal	The plausibility of the Zero Fuel quantity Calibration (ZFC) or Zero Fuel quantity calibration during Low idle (ZFL), used to correct pilot injection quantity is monitored for plausibility.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	60982-31
MoFMode1	ECU Internal: Diagnosis fault check to report the demand for normal mode due to an error in the PoI2 quantity	E1DD-1F	ECU Internal	The total fuel quantity contributing to torque generation is determined from the addition of pilot injections, main injection plus the product of post injection 2 and 3 and their efficiency (post injection 1 only used for DPF regeneration). Post injection 2 start angle and efficiency are evaluated here. Furthermore engine operation mode plausibility and mode transitions are monitored.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	57821-31
MoFMode2	ECU Internal: Diagnosis fault check to report the error to demand for an ICO due to an error in the PoI2 shut-off	E2DD-1F	ECU Internal	The total fuel quantity contributing to torque generation is determined from the addition of pilot injections, main injection plus the product of post injection 2 and 3 and their efficiency (post injection 1 only used for DPF regeneration). Post injection quantity 2 shut off in normal mode is evaluated.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	58077-31
MoFMode3	ECU Internal: Error in the plausibility of PoI3 efficiency.	E3DD-1F	ECU Internal	The total fuel quantity contributing to torque generation is determined from the addition of pilot injections, main injection plus the product of post injection 2 and 3 and their efficiency (post injection 1 only used for DPF regeneration). Post injection 3 efficiency plausibility is evaluated.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	58333-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoFOvR	ECU Internal: Diagnostic fault check to report the error due to Over Run	E4DD-1F	ECU Internal	The Actuation time of Common Rail injector is calculated by the demanded fuel injection quantity of actual injection part (Pilot, Main, Post...) and actual fuel rail pressure. As very low quantities must be guaranteed, there are correction factors evaluated, which compensate the tolerances and ageing behaviour of several components. This system is validated as "safety critical", because the injection duration can be increased by such corrections. Therefore the calculation of the final injector actuation time (calculated on level 1) is observed by a separate monitoring (level 2). If the actuation time during "Overrun" (engine driven by vehicle without combustion) exceeds a limit, the ECU is not working correctly. Note: If this failure appears, this means a fatal error in the ECU or a wrong data set calibration. Therefore contact helpdesk in this case.	ECM internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	58589-31
MoFQntCor	ECU Internal: Diagnostic fault check to report the error due to injection quantity correction	E5DD-1F	ECU Internal	The Common Rail system allows to have several injections (Pilot 1, Pilot 2, Main, Post 1, Post 2). Each injection part causes a pressure oscillation in the Rail, which can influence the quantity of the next injection. This is compensated by the "pressure wave correction", which adapts the "virtual requested" quantity of the actual injection part depending on the actual pressure situation, in the way, that the real injection quantity is equal to the desired one. The pressure wave correction is validated as "safety critical", because the procedure can increase the requested injection quantity. Therefore the calculation of the correction (Level 1) is observed by a second calculation (Level 2). If the result of the corrections on level 1 is not plausible with the result of Level 2, the ECU is not working correctly. Note: If this failure appears, this means a fatal error in the ECU or a wrong data set calibration. Therefore contact helpdesk in this case.	ECU internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	58845-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MoFRailP	ECU Internal: Diagnostic fault check to report the plausibility error in rail pressure monitoring	E6DD-1F	ECU Internal	The behaviour of Fuel pressure in Common Rail is a safety critical functionality. Therefore the correct functionality of the physical Fuel pressure monitoring (Level 1) is observed with an additional monitoring (level 2) to check, if the ECU is working correctly. If the Level 2 detects an implausible behaviour of the fuel pressure while the level 1 monitoring does not detect any inconvenience, the ECU is not working correctly. Note: If this failure appears, this means a fatal error in the ECU or a wrong data set calibration. Therefore contact helpdesk in this case.	ECU internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	59101-31
MoFTrqCmp	ECU Internal: Diagnostic fault check to report the error due to torque comparison	E7DD-1F	ECU Internal	The torque request can be influenced from a lot of vehicle functions (as accelerator pedal, ESP, ACS, automatic gear box etc) which are in some cases additive, in some cases as maximum and in some case as limitation. This complex structure can lead to a torque request, which is higher than accepted by engine. Therefore the total requested engine torque is compared to a maximal allowed engine torque, which is calculated by a redundant procedure. If the requested torque is higher, the system layout is not correct (calibration error in ECM) or the ECU does not work correctly. Therefore the ECU is reset (=switch off and automatic restart) 13.00- times to give the possibility to the system for correct calculation. If the requested torque is still higher than allowed, a fatal error in the system must be assumed, which is very critical. Therefore the engine speed is limited or in worst case the engine is stopped. Such intervention is confirmed with stored failure EngICO in failure memory. Note: if this failure happens, that means, that the ECU is not calibrated well or there is a fatal problem in the ECU. Contact the Help desk in case this failure is active.	ECU internal failure or calibration error.	Contact Helpdesk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	59357-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MonLimCurr	ECU Internal: Diagnosis of lead torque limitation (for fuel pressure control) forced by ECU monitoring level 2	E8DD-1F	ECU Internal	The torque request can be influenced from a lot of vehicle functions (as accelerator pedal, ESP, ACS, automatic gear box etc) which are in some cases additive, in some cases as maximum and in some case as limitation. Further the finally requested torque on engine must not be similar to the engine torque output, as the request takes into consideration system tolerances (e.g. due to learning factors, but also due temperatures and pressures), i.e. the final torque request is "virtual" in the way, that the real torque would be correct. As fuel pressure control is dependant on the torque input, the "virtual" torque request would lead to a wrong set point and therefore to wrong emissions. Therefore the input torque for the fuel pressure control is calculated separately - called "Lead torque". This complex structure can lead to a "lead" torque request, which is higher than accepted by engine. Therefore the total requested "lead" torque is compared to a maximal allowed (inner) torque. If the requested torque is higher, the system layout is not correct (calibration error in one of the electronic vehicle systems) and a failure is stored. Note: if this failure happens, that means, that one of the control systems is not calibrated well or there is a fatal problem in the ECU. Contact the Help desk in case this failure is active.	ECU internal failure or failure in the dataset calibration of ECU or another vehicle control unit	Contact help desk or Replace ECU	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	59613-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MonLimLead	ECU Internal: Diagnosis of lead torque limitation (for Air Control) forced by ECU monitoring level 2	E9DD-1F	ECU Internal	The torque request can be influenced from a lot of vehicle functions (as accelerator pedal, ESP, ACS, automatic gear box etc) which are in some cases additive, in some cases as maximum and in some case as limitation. Further the finally requested torque on engine must not be similar to the engine torque output, as the request takes into consideration system tolerances (e.g. due to learning factors, but also due temperatures and pressures), i.e. the final torque request is "virtual" in the way, that the real torque would be correct. As the air control (Boost pressure, EGR rate etc) is dependant on the torque input, the "virtual" torque request would lead to a wrong set point and therefore to wrong emissions. Therefore the input torque for the Air control is calculated separately - called "Lead torque". This complex structure can lead to a "lead" torque request, which is higher than accepted by engine. Therefore the total requested "lead" torque is compared to a maximal allowed (inner) torque. If the requested torque is higher, the system layout is not correct (calibration error in one of the electronic vehicle systems) and a failure is stored. Note: if this failure happens, that means, that one of the control systems is not calibrated well or there is a fatal problem in the ECU. Contact the Help desk in case this failure is active.	ECU internal failure or failure in the data set calibration of ECU or another vehicle control unit	Contact Help desk	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	59869-31
MonLimSet	ECU Internal: Diagnosis of engine torque limitation forced by ECU monitoring level 2	EADD-1F	ECU Internal	The torque request can be influenced from a lot of vehicle functions (as accelerator pedal, ESP, ACS, automatic gear box etc) which are in some cases additive, in some cases as maximum and in some case as limitation. This complex structure can lead to a torque request, which is higher than accepted by engine. Therefore the total requested engine torque is transferred to an "inner" engine torque (i.e. torque which has to be produced by the piston) and compared to a maximal allowed inner torque. If the requested torque is higher, the system layout is not correct (calibration error in one of the electronic vehicle systems) and a failure is stored. Note: if this failure happens, that means, that one of the control system is not calibrated well or there is a fatal problem in the ECU. Contact the Help desk in case this failure is active.	ECU internal failure or failure in the data set calibration of ECU or another vehcile cotnrol unit	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	60125-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
MonUMaxSupply1	ECU Voltage Supply: Reported OverVoltage of Supply	EBDD-1F	ECU Voltage Supply	The CJ945 is a chip with 18 power stage and an additional module to monitor ECM internal 5Volts supply. Voltage error of the internal 5 Volt supply is monitored.	Excessive voltage supply of a CJ945 component: High battery voltage, defective wiring, internal defect of the ECM	Check the battery for correct voltage supply, check wiring. If defect remains replace ECM (internal defect)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	60381-31
MonUMinSupply1	ECU Voltage Supply: Reported UnderVoltage of Supply	ECDD-1F	ECU Voltage Supply	The CJ945 is a chip with 18 power stage and an additional module to monitor ECM internal 5Volts supply. Voltage error of the internal 5 Volt supply is monitored.	Insufficient voltage supply of a CJ945 component: Low battery voltage, defective wiring, internal defect of the ECM	Check the battery for sufficient voltage supply, check wiring. If defect remains replace ECM (internal defect)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	60637-31
MRlyErlyOpng	ECU Main relay: Early opening defect of main relay	F561-1F	ECU Main relay	The Main relay is controlled by the ECU itself: After Key 15 off the ECU goes into the "After-run", where mainly following actions are made: - storage of the failures into the memory - counting of driving cycles and the corresponding reactions (pending --> validated failure, failure declared as long time failure etc.) Therefore the Main Relay is controlled by the ECU to assure the After-run. The main Relay is switched off by the ECU itself after the After-run has finished. This means, that all this actions can not be done, if the Main relay switches off before the After-run has finished. Therefore the correct opening of Main Relay is monitored. At each start a bit in the EEPROM is set, which is reset to 0 at the end of the After-run immediately before the ECU switches off the Main Relay. At next start the System checks the state of the bit in the EEPROM: If it is still set, this means that the Main Relay was switched off before the request of the ECU.	The Main Relay is switched off directly by Key 15 instead of the request of ECU: - in case vehicle equipped with "main electrical switch": engine stopped by "main switch" instead of Key 15 OR Main switch disconnected before afterrun finished. - due to wrong wiring to the Main Relay (direct contact to the battery missing or wiring ECU to Main relay) - due to internal defect of ECU	In case of vehicle equipped with "main electrical switch": inform driver of correct engine shut off by Key 15. In case of vehicle equipped with electronic "main electrical switch": check electronic of main switch for correct working (if opens after afterrun finished) Check wiring harness between main Relay and battery Check wiring harness between ECU and Main Relay Check ECU	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62817-31
NplPresSensCrCsP	Crankcase pressure: Value measured at ECM shut off not plausible	37B3-1F	Crankcase pressure	The plausibility of Crankcase differential pressure is evaluated during ECM after-run	Too high pressure in crankcase because of clogged Blow By Filter Too low pressure in crankcase because of clogged blow by filter	1 Open oil refill closure 2 Observe corresponding measuring channel CrCsP_pSens at engine stand still if the value is still out of the limit à3 3 Remove BlowBy Filter 4 Observe corresponding measuring channel CrCsP_pSens at engine stand still if the value is still out of the limit à5 5 component has a permant offsetà replace sensor	no lamp (OFF)	14259-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
OCWDACom	ECU Internal: Diagnostic fault check to report "WDA active" due to errors in query-/response communication	E81D-1F	ECU Internal	Communication between CJ945 (power stage chips) and CY320 (power stage voltage supplier) is monitored.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59421-31
OCWDALowVltg	ECU Internal: Diagnostic fault check to report "ABE active" due to undervoltage detection	E91D-1F	ECU Internal	Communication between CJ945 (power stage chips) and CY320 (power stage voltage supplier) is monitored.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59677-31
OCWDAOvrVltg	ECU Internal: Diagnostic fault check to report "ABE active" due to overvoltage detection	EA1D-1F	ECU Internal	Communication between CJ945 (power stage chips) and CY320 (power stage voltage supplier) is monitored.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	59933-31
OCWDAReasUnkwn	ECU Internal: Error report "ABE/WDA active" due to an unknown reason	EB1D-1F	ECU Internal	Communication between CJ945 (power stage chips) and CY320 (power stage voltage supplier) is monitored.	ECM internal failure	ECM need to be replaced, contact nearest head office for authorisation.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	60189-31
OilPSwmpPhysRngLo	Oil Pressure Sensor: Physical signal below minimum limit	26D3-1F	Oil pressure sensing	Oil pressure raw voltage value measured by sensor read from the ADC is converted to physical value Oil_pSwmp. Measured physical value should lie in a defined range.	Oil pressure too low, or inaccurate or defective Oil Pressure sensor, or defect in its wiring	Check oil level, check oil leakage, check wiring and the accuracy of Oil Pressure sensor and replace it if necessary	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	9939-31
OilPSwmpSRCMax	Oil pressure sensing: SRC high for oil pressure sensor	18D3-1F	Oil pressure sensing	Oil pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through analogic to Digital Converter and transformed to a physical value in Oil_pSensPSwmp. A signal range check is performed on the signal.	Sensor defective or wiring short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	6355-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
OilPSwmpSRCMin	Oil pressure sensing: SRC low for Oil pressure sensor	29D3-1F	Oil pressure sensing	Oil pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through analogic to Digital Converter and transformed to a physical value in Oil_pSensPSwmp. A signal range check is performed on the signal.	Sensor defective or short circuit to ground in the wiring	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	10707-31
OilTPhysRngHi	Oil Temperature sensing: Physical Signal above maximum limit	1BD3-1F	Oil Temperature sensing	Oil temperature is measured by a sensor connected via hardware pin to the ECM. Raw value is processed by analogic Digital Converter into physical oil temperature Oil_tSensSwmp. Measured physical value should lie in a defined range.	Excessive measured Oil temperature	Check cause for exceeding maximum allowed threshold (like environmental factors, system problem or engine misuse). If threshold exceeding is implausible check wiring and the accuracy of the sensor and replace it if necessary.	no lamp (OFF)	7123-31
OilTPhysRngLo	Oil Temperature sensing: Physical Signal below minimum limit	2CD3-1F	Oil Temperature sensing	Oil temperature is measured by a sensor connected via hardware pin to the ECM. Raw value is processed by analogic Digital Converter into physical oil temperature Oil_tSensSwmp. Measured physical value should lie in a defined range.	Excessively low measured Oil temperature	Check cause for value to fall below minimum allowed threshold (like environmental factors, system problem or engine misuse). If threshold exceeding is implausible check wiring and the accuracy of the sensor and replace it if necessary.	no lamp (OFF)	11475-31
OilTSRCMax	Oil Temperature sensing: SRC High for Oil Temperature	11E3-1F	Oil Temperature sensing	Oil temperature is measured by a sensor connected via hardware pin to the ECM. Raw value is processed by analogic Digital Converter into physical oil temperature Oil_tSensSwmp. A signal range check is performed on the signal.	Sensor defective or wiring short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4579-31
OilTSRCMin	Oil Temperature sensing: SRC low for Oil Temperature	22E3-1F	Oil Temperature sensing	Oil temperature is measured by a sensor connected via hardware pin to the ECM. Raw value is processed by analogic Digital Converter into physical oil temperature Oil_tSensSwmp. A signal range check is performed on the signal.	Sensor defective or wiring short circuit to ground	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	8931-31
PFuPreFilSRCMax	Fuel Pre-FIlter Pressure: Signal Range Check High	15E5-1F	Fuel pressure	Fuel Pre-Filter Pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through Analogic to Digital Converter and transformed to a physical value. A signal range check is performed on the signal.	Sensor defective or short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5605-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
PFuPreFilSRCMin	Fuel Pre-FIlter Pressure: Signal Range Check Low	16E5-1F	Fuel pressure	Fuel Pre-Filter Pressure sensor is available and connected via hardware to the ECM. Pressure is obtained through Analogic to Digital Converter and transformed to a physical value. A signal range check is performed on the signal.	Sensor defective or short circuit to ground	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5861-31
PhyModNonMonMapNpl	ECU internal Check: Not plausible fault: non strictly monotonous q curves quantity in conversion maps fuel quantity /torque	E1FD-1F	ECU internal Check	The map for the conversion from fuel quantity to torque is checked for consistency.	Wrong application of the basic map for conversion of injection quantity to inner torque and vice versa (FMTC_trq2qBas_MAP).	Application error, the dataset has a faulty application and should not be used.	no lamp (OFF)	57853-31
PIntkVUsSRCMax	Boost pressure sensor: SRC High Air Intake pressure value	1D64-1F	Boost pressure	The electrical Signal of the boost pressure sensor is monitored. In case of defect recognition a replacement value is taken.	Short circuit of sensor to external source or defective sensor	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	7524-31
PIntkVUsSRCMin	Boost pressure sensor: SRC Low Air Intake pressure value	2E64-1F	Boost pressure	The electrical Signal of the boost pressure sensor is monitored. In case of defect recognition a replacement value is taken.	Short circuit at wiring harness or inside the boost pressure sensor to ground or open circuit	Check of wiring or replace boost pressure sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	11876-31
PRVctOpnMax	Pressure Relief Valve: Opening count of PRV for wear out monitoring had exceeded maximum value	1145-1F	Pressure Relieve Valve	For determination of the mechanical state of the pressure relief valve it is detected, if the pressure relief valve is forced to open too many times.	PRV opens to often. Possible failure causes: Metering unit or wiring harness defective, rail pressure sensor failure, PRV problem (PRV opening pressure too low)	Check other rail failure, check PRV valve tightness and opening pressure, check metering unit, check wiring harness The PRV needs to be replaced and counter for number of opening should be reset using appropriate routine control service.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4421-31
PRVFrOpnPresInc	Pressure Relief Valve: Pressure shock test performed, first step, rail pressure increase	F245-1F	Pressure Relieve Valve	To determine mechanical wearout, the Pressure Relief Valve can be forced to open by performing a defined rail pressure shock in two steps: - Waiting for rail pressure build up - Performing pressure shock The two steps are recorded in the failure memory Pressure shock is only possible in a defined engine speed range depending from rail pressure.	Pressure increase and pressure shock are means to set system into limp home mode. In case of rail pressure sensor failure or maximum rail pressure exceeded, the system is set into limp home mode by performing pressure increase (MeUn set to full delivery) and pressure shock (MeUn set to full delivery and injection quantity set to zero)	No intervention necessary	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62021-31

Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
PRVFrOpnPresShck	Pressure Relief Valve: Pressure shock test performed, second step, rail pressure shock	F345-1F	Pressure Relieve Valve	To determine mechanical wearout, the Pressure Relief Valve can be forced to open by performing a defined rail pressure shock in two steps: - Waiting for rail pressure build up - Performing pressure shock The two steps are recorded in the failure memory Pressure shock is only possible in a defined engine speed range depending from rail pressure.	Pressure increase and pressure shock are means to set system into limp home mode. In case of rail pressure sensor failure or maximum rail pressure exceede, the system is set into limp home mode by performing pressure increase (MeUn set to full delivery) and pressure shock (MeUn set to full delivery and injection quantity set to zero)	No intervention necessary	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	62277-31
PRVOpn	Info: Pressure Relief Valve Opened	F445-1F	Pressure Relieve Valve	If rail pressure exceeds normal operation values a mechanical valve (PRV= pressure Relief Valve) is opened to control maximum rail pressure	Pressure Relief Valve is open due to: Failure of rail pressure sensor, wiring harness, metering unit or rail pressure failure	Check Rail pressure sensor, metering unit, wiring harness connection, rail system failure or PRV opening pressure decrease	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	62533-31
PRVtiOpnMax	Pressure Relief Valve: Open time of PRV for wear out monitoring had exceeded maximum value	1745-1F	Pressure Relieve Valve	The monitoring of the wear out of the PRV is done by counting the number of times the PRV is opened and the duration of its opening. The duration of opening (PRV_tiOpn_mp) is accumulated in every driving cycle and stored in the EEPROM in afterrun state of the engine. If the PRV was open for a longer time than the applied limit a failure is detected.	PRV opens to often. Possible failure causes: Metering unit or wiring harness defective, rail pressure sensor failure, PRV problem (PRV opening pressure too low)	Check other rail failure, check PRV valve tightness and opening pressure, check metering unit, check wiring harness, check RDS. The PRV needs to be replaced and timer monitoring opening status should be reset using appropriate routine control service.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5957-31
RailBSA	Injector Blank Shot Actuation aborted , rail pressure not released after engine stop	F895-1F	Injector	In rail system with only one actuator and injectors without recirculation, when engine is stopped rail cannot be depressurised (no system leakage present). For this reason a blank shot functionality was addede during afterrun. A blank shot refers to a very brief activation of the injector that causes the injector control quantity to be discharged in order to reduce the rail pressure, without injecting any fuel into the cylinder	Abnormal engine stop condition? Afterrun interrupted?	INFO failure: No intervention necessary	no lamp (OFF)	63637-31
RailLeakTst	Monitoring for pressure loss in the hig pressurel fuel rail	3995-1F	Fuel pressure (Rail) control	A leakage in the rail system causing fuel spilling into engine oil could cause a dangerous oil diluition. A monitor for unexpected rail pressure variation at engine stop and during overrun phases was introduced to detect possible rail leakages.	Leakage in the rail system present	Check rail system for leakage, check oil for diluition (if rail leakage is inside engine overhead). Repair leakage and replace engine oil if necessary	no lamp (OFF)	14741-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
RailMeUn0	Fuel pressure control Metering unit: Too low fuel rail pressure with MeUn Control	2265-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. If desired fuel pressure can not be reached, a leaky system must be assumed. Note: This Monitoring is only active with non empty fuel tank.	Leakage in the high pressure system: injection nozzle stuck in open position, low efficiency of high pressure pump, internal leakage of injector, leaking pressure control valve Too low pressure at High pressure pump inlet due clogged filter, leaking low pressure tubes, electric pre supply pump defective or gear pump defective	Check low pressure system for clogging (specially fuel filter) Check tubing between pre supply and gear pump Check feeding of pre supply pump (possibly low due to clogged tubes at tank outlet) Check high pressure part for leakages (internal / external) Check High pressure pump for efficiency Check injectors for tightness	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	8805-31
RailMeUn10	Fuel pressure (Rail) control: Leakage in high pressure system	4465-1F	Fuel pressure (Rail) control	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) as feeding quantity control The desired fuel flow is adjusted by feed back control for reaching the desired fuel pressure. If this needed fuel flow exeeds a maximal expected fuel flow by more than a fuel flow dependant value (6000.00mm3/s at 0.00mm3/s to 5800.00mm3/s at 16400.00mm3/s), leakage in the system must be assumed. Note: This Monitoring is only active with non empty fuel tank.	External leakage in the high pressure or low pressure system. Fuel pressure sensor signal shows wrong (mainly too low) values. Injectors leak internally (do not close) High pressure pump has too low feeding efficiency Restriction in fuel supply	Check if system has no external leakage Check if fuel pressure sensor signal is correct Check if injectors close well Check High pressure pump delivery Check fuel supply line and check for clogged filter	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	17509-31
RailMeUn2	Fuel pressure control Metering unit: Too high fuel pressure with MeUn on 0-delivery state	1565-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. If the desired fuel pressure is exceeded although the MeUn is fully closed (0-delivery) the system is not OK Note: This Monitoring is only active with non empty fuel tank.	Metering unit stuck in open position, zero-delivery throttle clogged, metering unit without power due to electrical error. Low pressure side: Pressure after zero-delivery throttle too high (Pump backflow pressure too high)	Check metering Unit (MeUn), if closing (PWM = 100%) Check electric wiring to MeUn Check low pressure fuel tubes (return from CP to tank free?) Check zero delivery valve for correct working	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	5477-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
RailMeUn3	Fuel pressure control Metering unit: Fuel rail pressure below limit with MeUn Control	2765-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. If the minimal desired fuel pressure is not reached, the system is not OK Note: This Monitoring is only active with non empty fuel tank.	Leakage in the high pressure system: injection nozzle stuck in open position, low efficiency of high pressure pump, internal leakage of injector, leaking pressure control valve Too low pressure at High pressure pump inlet due clogged filter, leaking low pressure tubes, electric pre supply pump defective or gear pump defective	Check low pressure system for clogging (specially fuel filter) Check tubing between pre supply and gear pump Check feeding of pre supply pump (possibly low due to clogged tubes at tank outlet) Check high pressure part for leakages (internal / external) Check High pressure pump for efficiency Check injectors for tightness	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	10085-31
RailMeUn4	Fuel pressure control Metering unit: Fuel Rail pressure above maximum limit with MeUn Control	1865-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. If the fuel pressure exceeds 2350.00bar, the system is not OK Note: This Monitoring is only active with non empty fuel tank.	Metering unit stuck in open position, zero-delivery throttle clogged, metering unit without power due to electrical error. Low pressure side: pressure after zero-delivery throttle too high (Pump backflow pressure too high)	Check metering Unit (MeUn), if closing (PWM = 100%) Check electric wiring to MeUn Check low pressure fuel tubes (return from CP to tank free?) Check zero delivery valve for correct working	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	6245-31
RailMeUn7	Fuel pressure control Metering unit: Leakage in overrun	3B65-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. Highpressure system leakage is checked at overrun. Note: This Monitoring is only active with non empty fuel tank.	Leakage in the high pressure system: low efficiency of high pressure pump, internal leakage of injector, leaking pressure control valve Too low pressure at High pressure pump inlet due clogged filter, leaking low pressure tubes, electric presupply pump defective or gear pump defective	Check low pressure system for clogging (specially fuel filter) Che tubing between presupply and gear pump Check feeding of presupply pump (possibly low due to clogged tubes at tank outlet) Check hgih pressure part for leakages (internal / external) Check High pressur pump for efficiency	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	15205-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
RailMeUn8	Fuel pressure control Metering unit: Leakage in high pressure system detected at idle	3C65-1F	Fuel pressure control Metering unit	The Fuel Rail pressure control is executed with a Metering Unit (MeUn) device as feeding quantity control. If the desired fuel quantity to reach the requested rail pressure is higher than expected at idle speed, a leaky system must be assumed. Note: This Monitoring is only active with non empty fuel tank.	Leakage in the high pressure system: injection nozzle stuck in open position, low efficiency of high pressure pump, internal leakage of injector, leaking pressure control valve Too low pressure at High pressure pump inlet due clogged filter, leaking low pressure tubes, electric presupply pump defective or gear pump defective	Check low pressure system for clogging (specially fuel filter) Che tubing between presupply and gear pump Check feeding of presupply pump (possibly low due to clogged tubes at tank outlet) Check hgih pressure part for leakages (internal / external) Check High pressur pump for efficiency Check injectors for tightness	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	15461-31
RailPGradMon	Fuel pressure (Rail) sensor: Rail pressure raw value is intermittent	4125-1F	Fuel pressure (Rail) sensor	A loose contact in the rail pressure sensor wiring can lead to incorrect measuring of the rail pressure. If this does occur, typically the measured pressure rises considerably for a brief time (jumps > 350.00bar in a 10 ms window). This kind of extreme pressure increase does not occur in the system. To determine a malfunction based on this appearance, gradient monitoring of the rail pressure is performed.	Periodical loose of contact in the weiring, the connectors or sensor itself	Check wiring and connectors of Rail pressure sensor for good contacts. Check sensor, replace sensor if necessary and reset learning values in EEPROM / start New learning	no lamp (OFF)	16677-31
RailPOfsTstMax	Fuel pressure (Rail) sensor: rail pressure raw value is above maximum offset	1225-1F	Fuel pressure (Rail) sensor	The physical Fuel Rail pressure is monitored against offset drift at engine stand still after a waiting period, which allows to drop the pressure in the rail to ambient pressure. In this condition, the fuel pressure must be similar to 1 bar absolute.	Rail pressure sensor signal inaccurate	Ensure correct wiring and proper functioning of rail pressure sensor, Replace sensor if necessary and reset learning values in EEPROM / start New learning	no lamp (OFF)	4645-31
RailPOfsTstMin	Fuel pressure (Rail) sensor: rail pressure raw value is below minimum offset	2325-1F	Fuel pressure (Rail) sensor	The physical Fuel Rail pressure is monitored against offset drift at engine stand still after a waiting period, which allows to drop the pressure in the rail to ambient pressure. In this condition, the fuel pressure must be similar to 1 bar absolute.	Rail pressure sensor signal inaccurate	Ensure correct wiring and proper functioning of rail pressure sensor, replace sensor Replace sensor if necessary and reset learning values in EEPROM / start New learning	no lamp (OFF)	8997-31
RailPRV4	Fuel pressure control Pressure Relief valve: maximum rail pressure in limp home mode with PRV	1D85-1F	Fuel pressure (Rail) control	This function is used for monitoring the rail pressure when the rail control loop is in limp home mode operated with a PRV.	PRV Valve stuck, fuel backflow clogged or fuel fuel low pressure cycle too high.	Check PRV Valve, check fuel backflow, check low fuel pressure level	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	7557-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
RailPSRCMax	Fuel pressure (Rail) sensor: Sensor voltage above upper limit	1425-1F	Fuel pressure (Rail) sensor	Electric monitoring of Fuel Rail Pressure sensor	Short circuit of wiring to external source or defective pressure sensor	Check wiring and proper functioning of rail pressure sensor Replace sensor if necessary and reset learning values in EEPROM / start New learning	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	5157-31
RailPSRCMin	Fuel pressure (Rail) sensor: Sensor voltage below lower limit	2525-1F	Fuel pressure (Rail) sensor	Electric monitoring of Fuel Rail Pressure sensor	Short circuit of wiring to ground or defective pressure sensor	Check wiring and proper functioning of rail pressure sensor Replace sensor if necessary and reset learning values in EEPROM / start New learning	Frequency: BLINK triggered Lamp: - Red Lamp (RSL) - Protected Lamp	9509-31
SSpMon1	ECU Power Stage 1: Error Sensor supply for sensors (if mounted): SCR pressure. HCI press. and temp., Acc.pedal 2, Boost press. and temp., Oil press.and temp., EGR diff.pressure	E22D-1F	ECU Power Stage 1	The hardware encapsulation of CY320 performs a diagnostic of the sensor supplies and monitors the communication Between processor and voltage supply device CY320. For reasons of safety the ECM provides 3 independent 5V voltage supplies for the sensors. These sensor supply voltages are monitored to ensure they remain within a given range. This check for voltage limits is done autonomously by the hardware (thresholds not applicable).	Excessive battery voltage, defect in wiring harness, electrical failure in connected sensors or in the ECM Involved pins are: 168,191,192,207,231 and 232	Check battery voltage, check wiring for correct voltage supply of the connected sensor and the ECM. Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	57901-31
SSpMon2	ECU Power Stage 2: Error Sensor supply for sensors (if mounted): Acc.pedal 1, fuel pre filter clogging, Fuel filter deltaP, Water pump clutch, EGR throttle, P3 sensor	E32D-1F	ECU Power Stage 2	The hardware encapsulation of CY320 performs a diagnostic of the sensor supplies and monitors the communication Between processor and voltage supply device CY320. For reasons of safety the ECM provides 3 independent 5V voltage supplies for the sensors. These sensor supply voltages are monitored to ensure they remain within a given range. This check for voltage limits is done autonomously by the hardware (thresholds not applicable).	Excessive battery voltage, defect in wiring harness, electrical failure in connected sensors or in the ECM Involved pins are: 166,167,208,209,210 and 223	Check battery voltage, check wiring for correct voltage supply of the connected sensor and the ECM. Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58157-31
SSpMon3	ECU Power Stage 3: Error Sensor supply for sensors (if mounted): Humidity sensor (temp.and humidity), Rail pressure, Throttle position	E42D-1F	ECU Power Stage 3	The hardware encapsulation of CY320 performs a diagnostic of the sensor supplies and monitors the communication Between processor and voltage supply device CY320. For reasons of safety the ECM provides 3 independent 5V voltage supplies for the sensors. These sensor supply voltages are monitored to ensure they remain within a given range. This check for voltage limits is done autonomously by the hardware (thresholds not applicable).	Excessive battery voltage, defect in wiring harness, electrical failure in connected sensors or in the ECM Involved pins are: 190,211,234	Check battery voltage, check wiring for correct voltage supply of the connected sensor and the ECM. Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58413-31
SSpMonExt1	Error sensors supplies voltage tracker	E52D-1F	ECU Voltage Supply	Error sensors supplies voltage tracker			no lamp (OFF)	58669-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
SSpMonSply1Max	Monitoring of 12V sensor supply 1 SRC_HIGH Error	E62D-1F	ECU Voltage Supply	All 12V supplies in the ECU are monitored against to high or low voltages dependent from the battery voltage.	Defect in wiring harness, electrical failure in connected sensors or in the ECM	Check wiring for correct voltage supply of the connected sensor.Check connected sensors electrical integrity. Replace ECM.	no lamp (OFF)	58925-31
SSpMonSply1Min	Monitoring of 12V sensor supply 1 SRC_LOW Error	E72D-1F	ECU Voltage Supply	All 12V supplies in the ECU are monitored against to high or low voltages dependent from the battery voltage.	Defect in wiring harness, electrical failure in connected sensors or in the ECM	Check wiring for correct voltage supply of the connected sensor.Check connected sensors electrical integrity. Replace ECM.	no lamp (OFF)	59181-31
SSpMonSplyMax	Monitoring of internal 12V supply SRC_HIGH Error	E82D-1F	ECU Voltage Supply	All 12V supplies in the ECU are monitored against to high or low voltages dependent from the battery voltage.	Defect in wiring harness, electrical failure in connected sensors or in the ECM	Check wiring for correct voltage supply of the connected sensor.Check connected sensors electrical integrity. Replace ECM.	no lamp (OFF)	59437-31
SSpMonSplyMin	Monitoring of internal 12V supply SRC_LOW Error	E92D-1F	ECU Voltage Supply	All 12V supplies in the ECU are monitored against to high or low voltages dependent from the battery voltage.	Defect in wiring harness, electrical failure in connected sensors or in the ECM	Check wiring for correct voltage supply of the connected sensor.Check connected sensors electrical integrity. Replace ECM.	no lamp (OFF)	59693-31
StrtHSSCB	Starter Relay HS: power stage output short circuit to battery	6722-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	Short circuit of wiring to external source or defective HS starter relay	Check of wiring or replacement of HS relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	26402-31
StrtHSSCG	Starter Relay HS: power stage output short circuit to ground	7822-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	Short circuit of wiring to ground or defective HS starter relay	Check of wiring or replacement of HS starter relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	30754-31
StrtLSOL	Starter Relay LS: power stage output No Load error	5922-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	Broken or disconnected wiring or defective LS starter relay	Check of wiring or replacement of LS starter relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	22818-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
StrtLSOvrTemp	Starter Relay LS: power stage output over temperature error	8A22-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	High battery voltage, high temperature inside ECU, high load or wiring problem	Check battery voltage, wiring, power stage and LS starter relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	35362-31
StrtLSSCB	Starter Relay LS: power stage output short circuit to battery.	6B22-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	Short circuit of wiring to external source or defective LS starter relay	Check of wiring or replacement of LS starter relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	27426-31
StrtLSSCG	Starter Relay LS: Power stage output short circuit to ground	7C22-1F	Starter control	Engine Starter is controlled by two relays actuators one for Low Side and one for High Side. Both power stages are monitored by Hardware for electric failures.	Short circuit of wiring to ground or defective LS starter relay	Check of wiring or replacement of LS starter relay	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	31778-31
SWReset_0	ECU SW Reset: Software Resets visible to all testers	ED1D-1F	ECU SW Reset	A software internal failure occurred which lead to a software reset. This reset caused a failure which is only visible for manufacturer specific testers.	Electronic disturbances, various hardware defects (ECU internal) or configuration problems	If error occurs repeatedly, reprogram EDC. If error remains, replace EDC.	no lamp (OFF)	60701-31
SWReset_1	ECU SW Reset: Software Resets, visible to manufacturer Tester only	EE1D-1F	ECU SW Reset	A software internal failure occurred which lead to a software reset. This reset caused a failure which is only visible for manufacturer specific testers.	Electronic disturbances, various hardware defects (ECU internal) or configuration problems	If error occurs repeatedly, reprogram EDC. If error remains, replace EDC.	no lamp (OFF)	60957-31
T50Err	ECU Key 16: Defective T50 switch	4761-1F	ECU Key 16	Terminal 50 is used to indicate request for engine cranking Signal is acquired via hardware pin and updated into message T50_stRaw. Duration of cranking request is monitored.	Too long cranking request, short circuit in ignition key module or in wiring	Check ignition key module and wiring	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	18273-31
TECUPhysRngHi_0	ECU Temperature Sensor: Physical Signal ECU Temperature first sensor above maximum limit	115D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a defined range.	Excessive ECU temperature measured	Check cause for exceeding maximum allowed threshold (like environmental factors) Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4445-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
TECUPhysRngHi_1	ECU Temperature Sensor: Physical Signal ECU Temperature second sensor above maximum limit	125D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a defined range.	Excessive ECU temperature measured	Check cause for exceeding maximum allowed threshold (like environmental factors) Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4701-31
TECUPhysRngLo_0	ECU Temperature Sensor: Physical Signal ECU Temperature first sensor below minimum limit	235D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a defined range.	Excessively low ECU temperature measured	Check cause for value to fall below minimum allowed threshold (like environmental factors) Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9053-31
TECUPhysRngLo_1	ECU Temperature Sensor: Physical Signal ECU Temperature second sensor below minimum limit	245D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a defined range.	Excessively low ECU temperature measured	Check cause for value to fall below minimum allowed threshold (like environmental factors) Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9309-31
TECUSPIErr	ECU Temperature sensing: SPI Error ECU temperature sensor (LM71)	E55D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. A signal check is performed on the signal of each digital temperature sensor (LM71) connected via SPI. The Temperature evaluation is made by Hardware and the result sent by the message "TECU_numSPI" to the controller. In normal operation the bit 0 and bit 1 are both set (=1). If one of these bits is not set (=0), this means that the Hardware has detected an error.	Temperature sensor defective (inside ECM)	Replace ECM	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	58717-31
TECUSRCMax	ECU Temperature Sensor: Signal above maximum limit	165D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a plausible range.	Inaccurate or defective temperature sensor, defect in the wiring	Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	5725-31
TECUSRCMin	Diagnostic Fault Check for Signal below minimum limit	275D-1F	ECU Temperature sensing	High temperatures can damage the ECM processor, one or more sensor can be provided to monitor ECM temperatures. The selected temperature is monitored and should lie in a plausible range.	Inaccurate or defective temperature sensor, defect in the wiring	Check wiring and the accuracy of the temperature sensor and replace it if necessary (Sensor is ECU internal, only ECU replacement possible)	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	10077-31

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Name	Title	DTC	Component	General Remark	Failure Cause	Failure Repair	SVS	DTC SAE (SPN-FMI)
TIntkVUsSRCMax	Boost temperature: SRC high for air intake temperature	1374-1F	Boost temperature	The electrical Signal of the air temperature sensor upstream engine Inlet Valve, is monitored. In case of defect recognition a substitute value is taken.	Short circuit of sensor to external source or defective sensor	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4980-31
TIntkVUsSRCMin	Boost temperature: SRC low for air intake temperature	2474-1F	Boost temperature	The electrical Signal of the air temperature sensor upstream engine Inlet Valve, is monitored. In case of defect recognition a substitute value is taken.	Short circuit of sensor to ground or defective sensor	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9332-31
TSEACo_TSC1_CHKCNT	Torque or speed control demand disturbed or discontinued	ED2D-1F	CAN Bus	TSEACo_TSC1 message is checked for reliability (message discontinued or disturbed). For this purpose frame is provided with a checksum (for detecting message integrity) and a rolling counter (for detecting missing frames)	Defect on ABS / ASR / ESP CAN controller, CAN cable problem.	Check correct functioning of the CAN controller. Check wiring.	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	60717-31
WtrCooltP1SRCMax	Primary Coolant Pressure Sensor: SRC high for pressure sensor	1383-1F	Coolant pressure	Primary coolant pressure sensor is connected to ECU via hardware pin. Primary coolant pressure raw voltage value measured by sensor read from the ADC is converted to physical value. A signal range check is performed on the signal.	Sensor defective or wiring short circuit to external source	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	4995-31
WtrCooltP1SRCMin	Primary Coolant Pressure Sensor: SRC low for pressure sensor	2483-1F	Coolant pressure	Primary coolant pressure sensor is connected to ECU via hardware pin. Primary coolant pressure raw voltage value measured by sensor read from the ADC is converted to physical value. A signal range check is performed on the signal.	Sensor defective or short circuit to ground in the wiring	Check wiring or replace sensor	Frequency: ON triggered Lamp: - Amber Lamp (AWL) - Protected Lamp	9347-31

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